

ACRME — CR & CRG Deployment Layout Reference

Document Control

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Purpose	Visual reference for CR & CRG deployment structure across all environments and geographies

1. Purpose

This document provides the **authoritative deployment layout** for Azure Capacity Reservations (CR) and Capacity Reservation Groups (CRG) across all ACRME-managed environments. It translates the normative requirements (CAP-023, OPS-006/C-12, REG-003, ENV-003, PLC-010/PLC-010a) into concrete deployment patterns with examples for each geography.

Use this document to:

- Understand the structural organization of CRGs per environment and region
 - See how the regional + per-AZ CRG structure is implemented
 - Trace the naming convention for RGs, CRGs, and subscriptions
 - Visualize the difference between 3-region (US) and 2-region (Europe/Australia/Asia Pacific/Middle East) geographies
 - Understand provider/consumer subscription relationships for shared reservations
-

2. Core Structural Principles

2.1 CRG Structure per Environment (CAP-023)

For **each environment** (Prod, CVAL/NonProd, DR) within a subscription and region:

```

One environment in one region
├── One regional (non-zonal) CRG    ← holds SKUs without zonal reservation sup-
port
└── One CRG per availability zone    ← holds zone-bound reservations

```

Key rules:

- **Per-AZ CRGs** enforce zone isolation (CAP-011) **structurally** — zone-1 capacity can never be counted in another zone
- **Regional CRG** holds only SKUs that don't support zonal reservations
- **Environments are never mixed within a CRG** (ENV-003)
- Each CRG sits inside a resource group following the OPS-006 naming convention

2.2 Naming Convention (OPS-006, C-12)

All managed resources follow deterministic, parseable patterns:

Resource	Pattern	Example
Resource Group	rg-odcr-<env>-<region>-<NN>	rg-odcr-prod-eus2-01
CRG	crg-<env>-<region>-<scope>	crg-pr-eus2-az1
Subscription	sub-<org>-<domain>-<purpose>-<NN>	sub-jda-cld-core-01

CRG scope tokens: scope ∈ {reg, az1, az2, az3}

- reg = regional (non-zonal) CRG
- az1, az2, az3 = per-availability-zone CRGs

Environment tokens: env ∈ {prod, pr, cval, cv, nonprod, np, dr}

- Production: prod or pr
- CVAL/NonProd: cval, cv, nonprod, or np
- DR: dr

2.3 Environment Separation Rules (ENV-003)

Hard separation constraints (enforced structurally via separate CRGs):

- ❌ Non-prod and prod **cannot** share capacity
- ❌ DR and prod **cannot** share capacity
- ✅ DR **may** share with non-prod (where co-located, PLC-010)

3. Deployment Layout by Geography

3.1 US — Three-Region Model (REG-003)

Distribution: For a US geography deployment, **select 3 distinct regions** and **distribute** Prod, CVAL, and DR across them. Each region can host any/all environments; the “3-region model” describes the deployment distribution strategy, not inherent region capabilities.

In-scope regions (current):

- West US 3 (westus3)
- Central US (centralus)
- Canada Central (canadacentral)
- East US 2 (eastus2) — restricted, exception-only

Example deployment distribution:

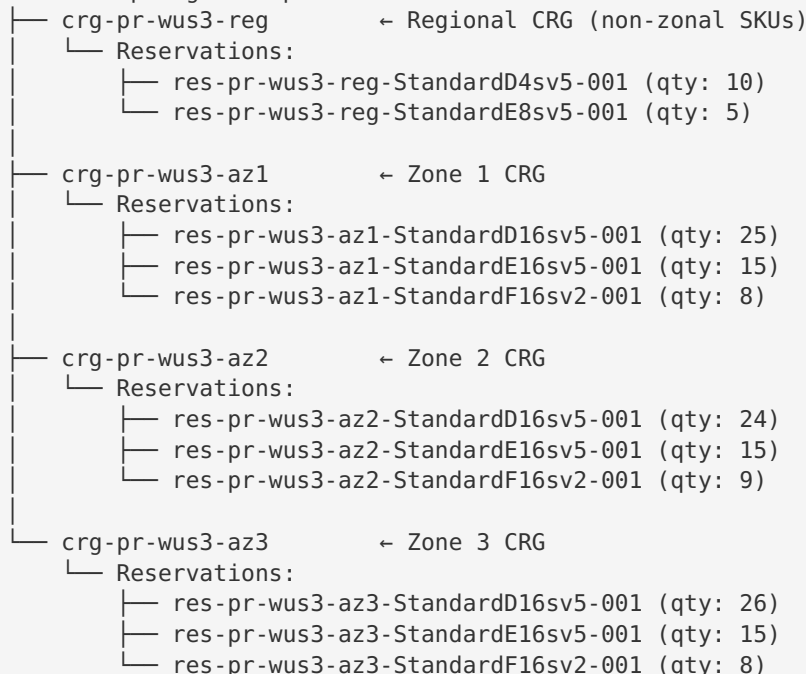
- **Prod** → **placed in:** West US 3
- **CVAL** → **placed in:** Central US
- **DR** → **placed in:** Canada Central

Note: Any of these regions can technically host all three environments simultaneously (each with separate subscriptions/RGs/CRGs per ENV-003). The distribution above shows a **deployment choice** for one customer, not a region capability limit.

3.1.1 Production Environment — West US 3 (3 zones)

Subscription: sub-jda-cld-core-01 (provider subscription, owns shared reservations)
 Region: West US 3 (westus3)
 Environment: Production

Resource Group: rg-odcr-prod-wus3-01



Sharing configuration (Tier 1/2):

Provider: sub-jda-cld-core-01

Consumers: sub-jda-cld-app-01, sub-jda-cld-app-02, ... (up to ~100)

Zone distribution (PLC-011):

- Target: ~33% per zone (1/3) in a 3-zone region
- Example: 75 total D16sv5 VMs → 25/24/26 distribution (within tolerance)
- Rebalance triggered when skew exceeds configured tolerance (C-13)

3.1.2 CVAL Environment — Central US (3 zones)

```

Subscription: sub-jda-cld-nonprod-01
Region: Central US (centralus)
Environment: CVAL/NonProd

Resource Group: rg-odcr-cval-cus-01
├── crg-cv-cus-reg          ← Regional CRG (non-zonal SKUs)
│   └── Reservations: (seed matrix, most at qty: 0)
├── crg-cv-cus-az1         ← Zone 1 CRG
│   └── Reservations:
│       ├── res-cv-cus-az1-StandardD8sv5-001 (qty: 5)
│       └── res-cv-cus-az1-StandardE8sv5-001 (qty: 3)
├── crg-cv-cus-az2         ← Zone 2 CRG
│   └── Reservations:
│       ├── res-cv-cus-az2-StandardD8sv5-001 (qty: 5)
│       └── res-cv-cus-az2-StandardE8sv5-001 (qty: 3)
└── crg-cv-cus-az3         ← Zone 3 CRG
    └── Reservations:
        ├── res-cv-cus-az3-StandardD8sv5-001 (qty: 5)
        └── res-cv-cus-az3-StandardE8sv5-001 (qty: 3)

```

CVAL characteristics:

- Separate subscription from Prod (ENV-003: prod ≠ non-prod)
- Lower buffer targets (C-2: configurable buffer policy)
- May be powered down for cost savings (ENV-002)
- In US 3-region model: CVAL is in a distinct region, **not co-located with DR**

3.1.3 DR Environment — Canada Central (3 zones)

```

Subscription: sub-jda-cld-dr-01
Region: Canada Central (canadacentral)
Environment: DR

Resource Group: rg-odcr-dr-cac-01
├── crg-dr-cac-reg          ← Regional CRG (non-zonal SKUs)
│   └── Reservations: (seed matrix, most at qty: 0 until DR event)
├── crg-dr-cac-az1         ← Zone 1 CRG
│   └── Reservations:
│       ├── res-dr-cac-az1-StandardD16sv5-001 (qty: 0) ← seed reservation
│       └── res-dr-cac-az1-StandardE16sv5-001 (qty: 2) ← bootstrap capacity
├── crg-dr-cac-az2         ← Zone 2 CRG
│   └── Reservations:
│       ├── res-dr-cac-az2-StandardD16sv5-001 (qty: 0) ← seed reservation
│       └── res-dr-cac-az2-StandardE16sv5-001 (qty: 2) ← bootstrap capacity
└── crg-dr-cac-az3         ← Zone 3 CRG
    └── Reservations:
        ├── res-dr-cac-az3-StandardD16sv5-001 (qty: 0) ← seed reservation
        └── res-dr-cac-az3-StandardE16sv5-001 (qty: 2) ← bootstrap capacity

```

DR characteristics (US 3-region):

- Separate region from Prod and CVAL
- Starts lean: bootstrap capacity only (DR-003, DR-004)
- Seed reservations at qty: 0 for eligible SKU/AZ matrix (CAP-022)
- Scales up on DR declaration (ADR-003, DR-009)
- Cannot share capacity with Prod (ENV-003), but **may** share with non-prod where policy permits

3.2 Europe — Two-Region Model with CVAL/DR Co-location (REG-003, PLC-010a)

Distribution: For a Europe geography deployment, **select 2 regions** and **distribute** environments: Prod in one region, CVAL + DR co-located in the other. Each region can host any/all environments; co-location is the deployment strategy for 2-region geographies.

In-scope regions (current):

- Switzerland North (switzerlandnorth) — authoritative cross-geo DR extension region (REG-002)
- Sweden Central (swedencentral)
- North Europe (northeurope) — restricted, exception-only
- West Europe (westeurope) — restricted, exception-only

Example deployment distribution:

- **Prod** → **placed in:** Sweden Central
- **CVAL + DR** → **co-located in:** Switzerland North

Note: Either region can technically host all three environments. The co-location pattern is required by PLC-010a for 2-region geographies (Prod isolation maintained; CVAL+DR share the second region to satisfy ENV-003 while delivering in-geo DR).

3.2.1 Production Environment — Sweden Central (3 zones)

Subscription: sub-jda-cld-core-eu-01
Region: Sweden Central (swedencentral)
Environment: Production

Resource Group: rg-odcr-prod-sdc-01

- ├─ crg-pr-sdc-reg ← Regional CRG
- ├─ crg-pr-sdc-az1 ← Zone 1 CRG
- ├─ crg-pr-sdc-az2 ← Zone 2 CRG
- └─ crg-pr-sdc-az3 ← Zone 3 CRG

(Structure identical to US Prod example above)

3.2.2 CVAL + DR Co-located — Switzerland North (3 zones)

CRITICAL: In a 2-region geography, once Prod is anchored, only one region remains → **CVAL and DR co-locate deterministically** (PLC-010a). This is the **normal, required outcome**, not an error state.

Subscription: sub-jda-cld-nonprod-eu-01 (owns CVAL reservations)
 sub-jda-cld-dr-eu-01 (owns DR reservations)
 Region: Switzerland North (switzerlandnorth)
 Environments: CVAL + DR (co-located in same region, separate subscriptions)

CVAL ENVIRONMENT

Resource Group: rg-odcr-cval-swn-01
 Subscription: sub-jda-cld-nonprod-eu-01

```

├─ crg-cv-swn-reg          ← CVAL Regional CRG
├─ crg-cv-swn-az1         ← CVAL Zone 1 CRG
│   └─ Reservations:
│       ├── res-cv-swn-az1-StandardD8sv5-001 (qty: 10)
│       └─ res-cv-swn-az1-StandardE8sv5-001 (qty: 6)
├─ crg-cv-swn-az2         ← CVAL Zone 2 CRG
│   └─ Reservations:
│       ├── res-cv-swn-az2-StandardD8sv5-001 (qty: 10)
│       └─ res-cv-swn-az2-StandardE8sv5-001 (qty: 6)
└─ crg-cv-swn-az3         ← CVAL Zone 3 CRG
    └─ Reservations:
        ├── res-cv-swn-az3-StandardD8sv5-001 (qty: 10)
        └─ res-cv-swn-az3-StandardE8sv5-001 (qty: 6)
  
```

DR ENVIRONMENT (co-located, separate subscription & CRGs)

Resource Group: rg-odcr-dr-swn-01
 Subscription: sub-jda-cld-dr-eu-01

```

├─ crg-dr-swn-reg          ← DR Regional CRG
├─ crg-dr-swn-az1         ← DR Zone 1 CRG
│   └─ Reservations:
│       ├── res-dr-swn-az1-StandardD16sv5-001 (qty: 0) ← seed
│       └─ res-dr-swn-az1-StandardE16sv5-001 (qty: 2) ← bootstrap
├─ crg-dr-swn-az2         ← DR Zone 2 CRG
│   └─ Reservations:
│       ├── res-dr-swn-az2-StandardD16sv5-001 (qty: 0) ← seed
│       └─ res-dr-swn-az2-StandardE16sv5-001 (qty: 2) ← bootstrap
└─ crg-dr-swn-az3         ← DR Zone 3 CRG
    └─ Reservations:
        ├── res-dr-swn-az3-StandardD16sv5-001 (qty: 0) ← seed
        └─ res-dr-swn-az3-StandardE16sv5-001 (qty: 2) ← bootstrap
  
```

Co-location accounting (HC-6, HC-7, PLC-010):

- Seed record: `cval_region == dr_region` (Switzerland North) + co-location flag set
- **CVAL capacity is earmarked as releasable** toward DR activation (DR-005/DR-006)
- **Never double-counted** as both live CVAL and available DR headroom
- Combined-capacity checks: `DR_floor ≤ (CVAL_allocated + DR_allocated)` in the co-located region
- On DR declaration: CVAL may be shut down, disassociated, or reassigned per runbook

Why co-location is mandatory here:

- Geography has only 2 Standard regions (Sweden Central, Switzerland North)
 - Prod anchored in Sweden Central → only Switzerland North remains
 - Cannot separate all three environments → CVAL + DR co-locate deterministically
 - This is the **normative outcome for any 2-region geography**, not an exception
-

3.3 Australia — Two-Region Model (same pattern as Europe)

In-scope regions:

- Australia East (australiaeast)
- Australia Southeast (australiasoutheast)

Example deployment distribution:

- **Prod** → **placed in:** Australia East
- **CVAL + DR** → **co-located in:** Australia Southeast

Note: Either region can host all three environments. Distribution follows the same 2-region co-location pattern as Europe.

Structure: Identical to Europe example above (§3.2), substitute region codes.

3.4 Asia Pacific — Two-Region Model (same pattern as Europe)

In-scope regions:

- East Asia (eastasia)
- Southeast Asia (southeastasia)
- Japan East (japaneast) — **pending business confirmation before inclusion**

Example deployment distribution (current 2-region):

- **Prod** → **placed in:** East Asia
- **CVAL + DR** → **co-located in:** Southeast Asia

Note: Either region can host all three environments. Distribution follows the same 2-region co-location pattern as Europe.

Structure: Identical to Europe example above (§3.2), substitute region codes.

3.5 Middle East — Two-Region Model, `DR_NOT_OFFERED` (DEC-001, DR-014)

In-scope regions:

- Saudi Arabia Central (saudiarabiacentral)
- UAE North (uaenorth)

Legal constraint: DR strategy pending legal confirmation; data-sovereignty requirements mean cross-border DR cannot meet residency rules. Current status: `DR_NOT_OFFERED` (DEC-001).

Example deployment distribution:

- **Prod** → **placed in:** Saudi Arabia Central
- **CVAL** → **placed in:** UAE North (or co-located with Prod)
- **DR:** ❌ Not offered (no DR region assigned; seed record: `dr_region = null`, `DR_NOT_OFFERED` flag set)

Note: Both regions can technically host Prod and CVAL. DR environment CRGs are **not built** for Middle East due to legal constraints (DEC-001), unlike other 2-region geographies where DR is offered via co-location.

PRODUCTION ENVIRONMENT – Saudi Arabia Central

Subscription: sub-jda-cld-core-me-01

Region: Saudi Arabia Central (saudiarabiacentral)

Resource Group: rg-odcr-prod-sac-01

```
├─ crg-pr-sac-reg
├─ crg-pr-sac-az1
├─ crg-pr-sac-az2
└─ crg-pr-sac-az3
```

CVAL ENVIRONMENT – UAE North (optional, if non-prod is offered)

Subscription: sub-jda-cld-nonprod-me-01

Region: UAE North (uaenorth)

Resource Group: rg-odcr-cval-uan-01

```
├─ crg-cv-uan-reg
├─ crg-cv-uan-az1
├─ crg-cv-uan-az2
└─ crg-cv-uan-az3
```

DR ENVIRONMENT – ❌ Not offered

No DR CRGs created.

Seed record: dr_region = null, DR_NOT_OFFERED = true

When legal approves DR for Middle East:

- Update region catalogue configuration (REG-001): remove `DR_NOT_OFFERED` flag
- DR will then co-locate with CVAL per the 2-region model (PLC-010a)
- Engine will auto-derive DR region = UAE North (or whichever region is not Prod)

4. Seed Matrix & Zero-Capacity Reservations (CAP-022)

Every **eligible SKU/family × region × availability zone** combination is initialized as a **seed reservation at quantity 0** inside the correct CRG, so reconciliation can scale it up instantly when demand appears.

Example seed matrix (partial) — Prod West US 3:

CRG	Reservation Name	SKU	Initial Qty	Product Team	Budget Line	State
crg-pr-wus3-az1	res-pr-wus3-az1-Standard-D2sv5-001	Standard_D2s_v5	0	Platform-Core	FY27-Q1-Compute-001	Seed
crg-pr-wus3-az1	res-pr-wus3-az1-Standard-D4sv5-001	Standard_D4s_v5	0	Platform-Core	FY27-Q1-Compute-001	Seed
crg-pr-wus3-az1	res-pr-wus3-az1-Standard-D8sv5-001	Standard_D8s_v5	0	Platform-Core	FY27-Q1-Compute-001	Seed
crg-pr-wus3-az1	res-pr-wus3-az1-Standard-D16sv5-001	Standard_D16s_v5	25	Platform-Core	FY27-Q1-Compute-001	Active (scaled from seed)
crg-pr-wus3-az1	res-pr-wus3-az1-Standard-E16sv5-001	Standard_E16s_v5	15	Data-Services	FY27-Q1-Compute-002	Active (scaled from seed)
crg-pr-wus3-az1	res-pr-wus3-az1-Standard-F16sv2-001	Standard_F16s_v2	8	ML-Platform	FY27-Q1-Compute-003	Active (scaled from seed)
...	...	...	...	...	...	...

Seed matrix governance (CAP-022):

- Entry requires: named owning **product team** + approved **budget line**
 - Anything above qty: 0 incurs cost → budget approval gates expansion
 - Matrix is configuration-driven (scope file, CAP-019) and versioned
 - Reactive discovery (CAP-024): unmanaged allocated SKU/AZ → auto-create reservation + raise scope-file governance item
-

5. Cross-Subscription Sharing (Tier 1/2/3)

5.1 Provider/Consumer Model (CAP-013, CAP-014, CAP-015)

Tier 1: Single subscription owns and consumes its own reservations (no sharing).

Tier 2 (Preview): Provider subscription shares reservations with consumer subscriptions within the same region + AZ (up to ~100 consumers per tenant).

Tier 3 (Preview): Provider subscription can forcibly disassociate VMs from consumer subscriptions (requires G-14 consumer credential model — currently undesigned).

5.2 Sharing Layout Example — Prod West US 3, Zone 1

PROVIDER SUBSCRIPTION (owns the shared reservation)

Subscription: sub-jda-cld-core-01
 Region: West US 3, Zone 1
 CRG: crg-pr-wus3-az1

Reservation: res-pr-wus3-az1-StandardD16sv5-001
 Reserved Quantity: 100
 Provider: sub-jda-cld-core-01
 Sharing enabled: YES
 Authorized consumers:
 - sub-jda-cld-app-01
 - sub-jda-cld-app-02
 - sub-jda-cld-app-03
 - ... (up to ~100)

CONSUMER SUBSCRIPTIONS (consume shared reservation, don't own it)

Subscription: sub-jda-cld-app-01
 Allocated VMs consuming shared reservation:
 - 15 × Standard_D16s_v5 in Zone 1 (associated to provider's CRG)
 Quota required: YES (consumer subscription must have its own quota, QUA-013)

Subscription: sub-jda-cld-app-02
 Allocated VMs consuming shared reservation:
 - 22 × Standard_D16s_v5 in Zone 1 (associated to provider's CRG)

Subscription: sub-jda-cld-app-03
 Allocated VMs consuming shared reservation:
 - 18 × Standard_D16s_v5 in Zone 1 (associated to provider's CRG)

TOTAL CONSUMPTION ACCOUNTING (CAP-015 – no double counting)

Provider reserves: 100
 Consumer-1 allocated: 15
 Consumer-2 allocated: 22
 Consumer-3 allocated: 18

Total allocated: 55 (counted once at provider)
 Remaining: 45 (available for new allocations)

Sharing constraints:

- Same **region AND availability zone** only (CAP-011, CAP-012)
- Authorization to consume ≠ allocated capacity (CAP-015)
- Each consumer needs its own quota (QUA-013)

- Sharing state tracked: provider, authorized consumers, consuming VMs, consumed/remaining quantities (CAP-014)

6. Reconciliation & Lifecycle

6.1 Target Formula (CAP-003)

Target Reserved Capacity = Allocated (running) VM Count + Configured Buffer

- **Allocated** = VMs currently consuming compute (running state)
- **Associated** (deallocated) VMs do **not** drive target (CAP-004)
- **Buffer** = configurable per environment (C-2: production buffer, DR buffer)

6.2 Reconciliation Loop (CAP-005, CAP-006)

```
Every 6 minutes (configurable, CAP-006):
  FOR each managed CRG:
    1. Query Azure: allocated VMs, associated VMs, reserved quantity
    2. Compute: target = allocated + buffer
    3. IF reserved < target:
      → Scale-up (CAP-007): attempt increase; alert if Azure cannot supply
    4. ELSE IF reserved > target:
      → Scale-down (CAP-008): reduce toward target (right-sizing retained reservation)
    5. Validate: quota sufficient for reserved capacity (QUA-007)
    6. Log: audit event (all changes tracked)
```

Scale-down discipline (CAP-008, CAP-009, CAP-010):

- Scale-down is **right-sizing** → reduce reservation toward `allocated + buffer`
- May reach qty: 0 if allocated legitimately drops to 0 (normal for seed reservations)
- **Never deletes** CRG or reservation object
- Intentional retirement/deletion → requires **separate decommissioning workflow** (CAP-010): approval + impact analysis + audit

6.3 Reactive Discovery (CAP-024)

```
IF (new allocated VM detected for SKU/AZ not in seed matrix):
  1. Auto-create CRG (if absent) + reservation at qty: (allocated + buffer)
    → Protects production immediately
  2. Raise scope-file governance item (CAP-019)
    → Product team + budget ratification required
  3. Scope file updated within governance SLA
    → Reactive addition becomes authoritative (or decommissioned if rejected)
```

7. Quick Reference Tables

7.1 CRG Count per Geography

Geo- graphy	Distribu- tion Mod- el	Prod Re- gions	CVAL Re- gions	DR Re- gions	CRGs per Env	Total CRGs (3 envs)
US	3-region	1	1 (distinct)	1 (distinct)	4 (1 reg + 3 AZ)	12
Europe	2-region	1	1 (co- located with DR)	1 (co- located with CVAL)	4 per env	8 (Prod: 4, CVAL+DR co-located: 4+4 same region)
Australia	2-region	1	1 (co- located with DR)	1 (co- located with CVAL)	4 per env	8
Asia Pa- cific	2-region	1	1 (co- located with DR)	1 (co- located with CVAL)	4 per env	8
Middle East	2-region	1	1 (option- al)	✗ Not offered	4 (Prod), 4 (CVAL)	8 (or 4 if CVAL not deployed)

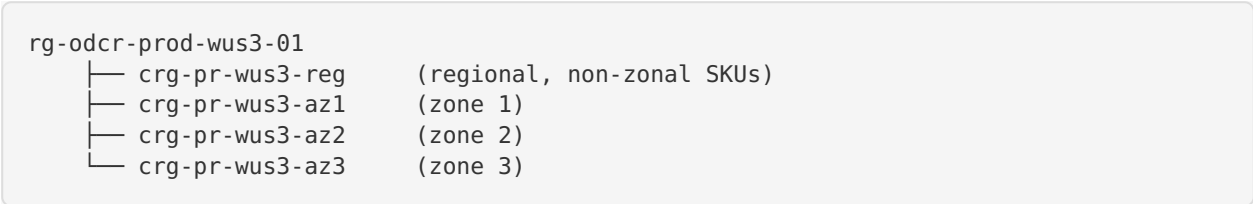
Note: 4 CRGs per environment assumes 3 availability zones per region (1 regional + 3 per-AZ). Adjust for regions with different zone counts.

7.2 Environment Separation Matrix (ENV-003)

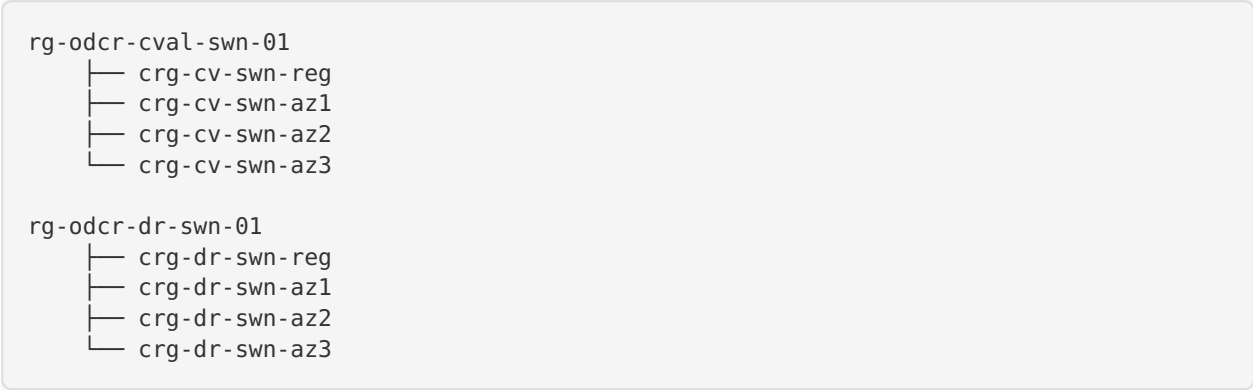
	Prod	CVAL/NonProd	DR
Prod	—	✗ Cannot share	✗ Cannot share
CVAL/NonProd	✗ Cannot share	—	✓ May share (where co-located)
DR	✗ Cannot share	✓ May share	—

7.3 Naming Pattern Quick Reference

3-zone region example (West US 3, Production):



2-region co-located example (Switzerland North, CVAL + DR):



8. Related Documents

- `acrme_requirements_baseline_v2_4.md` — normative source (CAP-001..024, OPS-006, REG-001..003, ENV-003, PLC-010/010a)
- `acrme_technical_design_document.md` — TDD service/schema detail
- `acrme_hard_constraints_reference.md` — HC-1..11 (zone isolation, DR coverage floor, etc.)
- `acrme_calculation_logic_reference.md` — 15-scenario calc logic, zone distribution formula (A.9)
- `acrme_security_and_rbac_guide.md` — custom roles, least-privilege UAMI model
- `Architecture/adr/acrme_adr_002_quota_management.md` — quota two-group model (ADR-002)
- `Architecture/adr/acrme_adr_003_capacity_management_during_dr.md` — DR capacity scaling (ADR-003)

9. Revision History

Version	Date	Change Summary
1.0	9 Sep 2026	Initial deployment layout reference; all geographies (US 3-region, Europe/Australia/Asia Pacific 2-region co-located, Middle East DR_NOT_OFFERED); CRG structure per CAP-023; naming per OPS-006; sharing model; seed matrix; reconciliation lifecycle.

End of document.